

Course Syllabus for 2110531

1. Course Number: 2110531
2. Course Credit: 3 (3-0-6)
3. Course Title: Data Science and Data Engineering Tools
4. Department: Computer Engineering
5. Semester: First Semester
6. Academic Year: 2026
7. Instructor: Prof. Peerapon Vateekul, Ph.D.
Asst. Prof. Natawut Nupairoj, Ph.D.
Assoc. Prof. Veera Muangsin, Ph.D.
8. Condition: -
9. Status: Elective
10. Curriculum: Computer Engineering
11. Degree: M.Eng., M.Sc.
12. Hours/Week: 3-Hour Lecture & Lab

13. Course Description

Data Science is the study of the discovery of knowledge from data. Being a data scientist requires an integrated skill set spanning mathematics, statistics, machine learning, databases, and other branches of computer science, along with a good understanding of the craft of problem solving. Data Engineering is the study of how to engineer or process data, i.e., data cleansing, data storing, etc. There are three main parts in this course:

- Data analysis: Machine Learning techniques
- Data engineering: Data exploration & preparation
- Data visualization: Storytelling via data

14. Course Outline

14.1. Learning Objectives

- Describe what Data Science and Data Engineering are and the skill sets needed
- To be able to explore and understand collected data
- To be able to analyze data by applying traditional machine learning techniques
- To be able to visualize data in relation to spatial and temporal points of view

14.2. Learning Contents

- Section 1 on Tue 1PM-4PM, Building ENG3 Room 417
- Section 5 on Sat 1PM-4PM, Building ENG4 Room 18-16
- *** Please bring your laptop to the class ***
- Students need to attend the class on-site at least 80% (at least 12 weeks) as a mandatory criterion to “pass” this course.

#	Tue (1PM-4PM)	Sat (1PM-4PM)	Topics	Professor	Module
1	04-Aug-26	08-Aug-26	Introduction, Pandas, Data Prep + AI-Assisted	Aj.Peerapon	DS1
2	11-Aug-26	15-Aug-26	Traditional ML (1)	Aj.Peerapon	DS2
3	18-Aug-26	22-Aug-26	Traditional ML (2)	Aj.Peerapon	DS3
4	25-Aug-26	29-Aug-26	Deep Learning (1); CNN, RNN (LSTM, GRU)	Aj.Peerapon	DS4
5	01-Sep-26	05-Sep-26	Deep Learning (2); Transformer Model monitoring (MLflow) Hyperparameter Tuning (Optuna)	Aj.Peerapon	DS5
6	08-Sep-26	12-Sep-26	Advanced topics (Prompt Engineering, RAG, Agentic AI); Tools: LangChain, LangGraph	Aj.Peerapon	DS6
7	15-Sep-26	19-Sep-26	Guest speaker + Midterm Announcement	Aj.Peerapon	DS7
	22-Sep-26	26-Sep-26	Midterm Exam Week (21 - 25 Sep 2026)		
8	29-Sep-26	03-Oct-26	Data visualization (Graduation Week 29 Sep - 1 Oct)	Aj.Veera	VIZ1
9	06-Oct-26	10-Oct-26	Data Storage (Data Lakehouse; Open Table Format) + Vector Database	Aj.Natawut	DE1
10	13-Oct-26	17-Oct-26	Web Scraping (LLM) + Data ingestion (Kafka) Note: 13-Oct-26 is King Bhumibol Adulyadej the Great Memorial Day	Aj.Natawut	DE2
11	20-Oct-26	24-Oct-26	Big data processing (Spark)	Aj.Natawut	DE3
12	27-Oct-26	31-Oct-26	MLOps: Orchestration (Airflow) and serving (FastAPI, Seldon Core)	Aj.Natawut	DE4
13	03-Nov-26	07-Nov-26	Spatial analysis	Aj.Veera	VIZ2
14	10-Nov-26	14-Nov-26	Graph analysis AI-Assited Data Analytics	Aj.Veera	VIZ3
15	17-Nov-26	21-Nov-26	MCP, AI Integration, Harness Engineering, Evaluation, Guardrails, Observability	Aj.Natawut	DE5
	24-Nov-26	28-Nov-26	Final Exam Week (23 Nov - 4 Dec 2026) *** Sat 28 Nov 2026 at 1PM-4PM ***		

14.3. Method: Lecture and Lab

14.4. Learning Media: PowerPoint presentation, Zoom

14.5. Evaluation

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| ○ Module1 Assignment (data analytics) | 15% |
| ○ Module2 Assignment (data engineering) | 15% |
| ○ Module3 Assignment (data visualization) | 5% |
| ○ Midterm Exam (Kaggle) | 15% |
| ○ Project | 15% |
| ○ Attendance | 5% |
| ○ Final Exam | 30% (Lab Test) |

15. Reading List

15.1. Required Text: N/A

15.2. Electronic Media or Websites: N/A

16. LMS

16.1. Github: https://github.com/pvateekul/2110531_DSDE_2026s1

16.2 MyCourseVille:

- <https://www.mycourseville.com/?q=courseville/course/86507> (Sec 1 Tue),
- <https://www.mycourseville.com/?q=courseville/course/86508> (Sec 5 Sat)

16.2. Discord: <https://discord.gg/MYvQtpV96H>

16.3. Grader: <https://dsde.nattee.net/>